Phase separation in heterogenous media

Joint work with Riccardo Cristoferi

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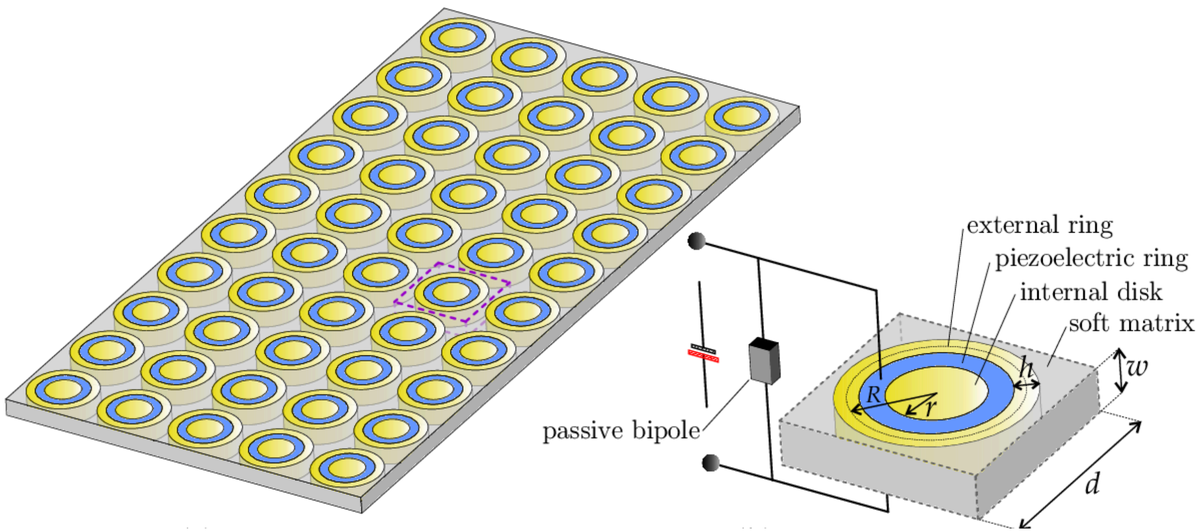
Motivation: Recent rise of metamaterials with periodic structure



Active acoustic metamaterial - Bacigalupo, Bellis, Misseroni (2019)

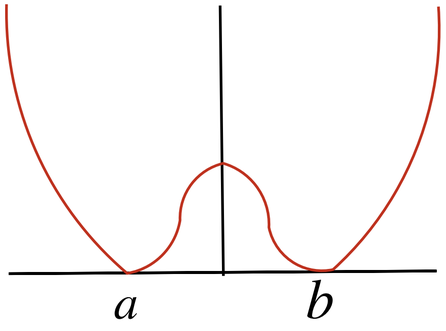
Metamaterial: a synthetic composite material with a structure such that it exhibits properties not usually found in natural materials: i.e.

- Negative refractive index
- Negative Poisson ratio

Objective: study stable equilibrium configurations of phase separation acting on these materials

Part 1: Phase separation in homogeneous media

Restricted to fixed two phases $\{a, b\}$, what to expect?

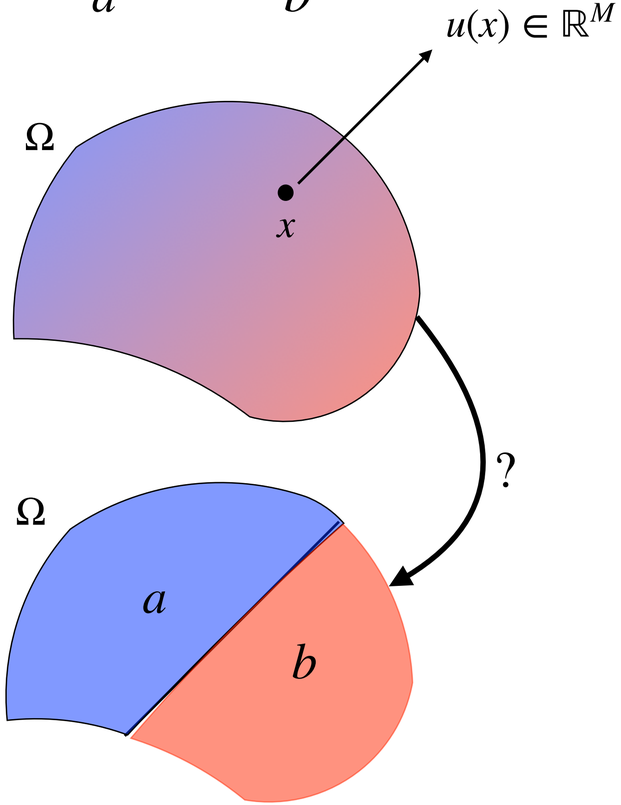


First idea: $W\colon \mathbb{R}^M \to [0, +\infty)$ continuous double well potential

$u\colon \Omega \to \mathbb{R}^M$ phase variable (immiscible liquids)

Minimise potential energy $F(u) = \int_\Omega W(u)\,\mathrm{d}x$ among $u \in L^1(\Omega; \mathbb{R}^M)$

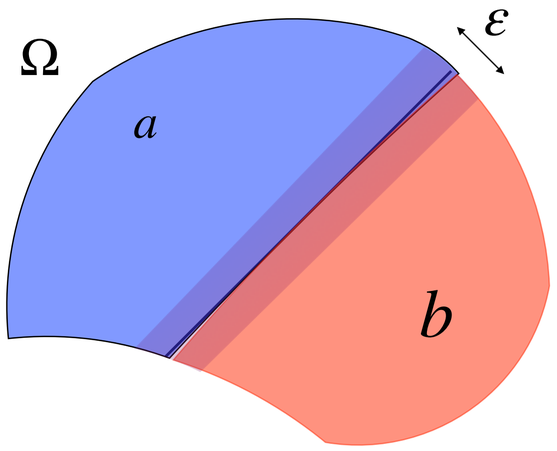
maybe subject to $\int_\Omega u\,\mathrm{d}x = m$



$\Longrightarrow$ solutions $u^* \in L^1(\Omega; \{a, b\})$, too wild to make sense of

Second idea: turn to a physical based model $\Longrightarrow$ Van der Waals (1893), Cahn-Hilliard (1958)

Ingredients:
- ε order parameter
- $|\nabla u|^2$ singular perturbation
- Gibbs free energy from thermodynamics



Minimize $F_\varepsilon(u) = \int_\Omega \left[W(u) + \varepsilon^2 |\nabla u|^2 \right] \mathrm{d}x$ over $u \in W^{1,2}(\Omega; \mathbb{R}^M)$

Solution will be characterised by a transition layer of size δ, physical parameter

$$F_\varepsilon(u) \approx c\delta + \varepsilon^2 \frac{1}{\delta^2}\delta \quad \text{which is minimised when} \quad \delta \approx \varepsilon$$

What happens when $\varepsilon \to 0$?

$$F_\varepsilon(u) \xrightarrow{w-\Gamma(L^2)} F^{(0)}(u) = \int_\Omega W(u)\,\mathrm{d}x$$

Not that interesting…

$$\min \left\{ F^{(0)}(u) : u \in L^1(\Omega; \mathbb{R}^M) \right\} = 0$$

Asymptotic Γ-expansion

Anzellotti-Baldo (1993)

Selection criterion for minimisers of F_ε

$$\varepsilon \ll 1 \quad \Longrightarrow \quad F_\varepsilon \approx F^{(0)} + \omega_\varepsilon^{(1)} F^{(1)} + \omega_\varepsilon^{(2)} F^{(2)} + \cdots$$

such that $\min F_\varepsilon \approx \min F^{(0)} + \omega_\varepsilon^{(1)} \min F^{(1)} + \omega_\varepsilon^{(2)} \min F^{(2)} + \cdots$

$$F_\varepsilon \xrightarrow{\Gamma} F^{(0)} \qquad F_\varepsilon^{(1)} = \frac{F_\varepsilon - \min F^{(0)}}{\omega_\varepsilon^{(1)}} \xrightarrow{\Gamma} F^{(1)} \qquad F_\varepsilon^{(2)} = \frac{F_\varepsilon - \min F^{(0)} - \omega_\varepsilon^{(1)} \min F^{(1)}}{\omega_\varepsilon^{(2)}} \xrightarrow{\Gamma} F^{(2)} \quad \cdots$$

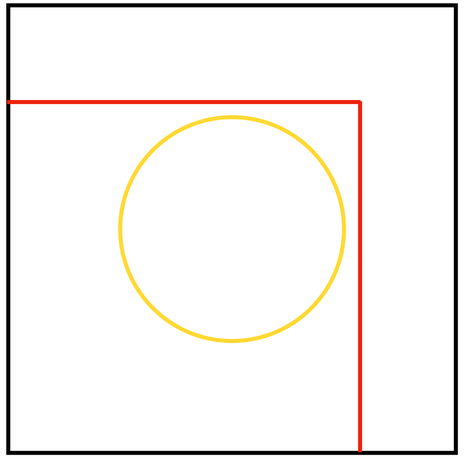
Drawbacks:

- Computing $F^{(i)}$ requires computing $\min F^{(j)}$ for $j = 1, \ldots, i-1$
- Finding the appropriate scaling $\omega_\varepsilon^{(i)}$ is mostly heuristic and luck
- At each stage, the existence of the Γ-limit is not guaranteed

Applied to our functional:

Zeroth order: $F_\varepsilon \xrightarrow{w-\Gamma(L^2)} F^{(0)}(u) = \int_\Omega W(u)\,\mathrm{d}x \qquad \min F^{(0)} = 0 \qquad \omega_\varepsilon^{(1)} = \varepsilon$

First order: $F_\varepsilon^{(1)}(u) = \int_\Omega \left[\frac{1}{\varepsilon}W(u) + \varepsilon\,|\nabla u|^2\right]\mathrm{d}x$ Modica (1986)



Two minimisers for the perimeter

$$F_\varepsilon^{(1)} \xrightarrow{\Gamma(L^1)} F^{(1)}(u) = \sigma\,\mathrm{Per}(\{u = a\};\Omega) \qquad u \in \mathrm{BV}(\Omega;\{a,b\}) \qquad \min F^{(1)} \neq 0 \qquad \omega_\varepsilon^{(2)} = \varepsilon^2$$

$$\sigma = \inf\left\{\int_{-1}^{1} 2\sqrt{W(\gamma(t))}\,|\gamma'(t)|\,\mathrm{d}t : \gamma \in \mathrm{Lip}([-1,1];\mathbb{R}^M),\ \gamma(-1) = a,\ \gamma(1) = b\right\}$$

σ is the “minimal energy" to change phase, achieved with optimal profile

FYI

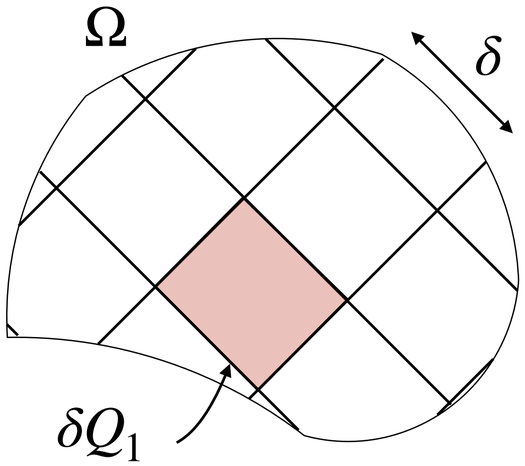
Second order: $F^{(2)}(u) = c_1\,\kappa_u^2 + c_2\,\kappa_u\,\mathrm{Per}(\{u = a\};\Omega)$

where κ_u is the mean curvature on the boundary of $\{u = a\}$ Leoni, Murray (2016)

Part 2: Phase separation in heterogeneous media

Which heterogeneities? Multiple periodic microstructures, nested and possibly with inclusions

Microstructure present only in potential



One-scale microstructure: $$F_{\varepsilon,\delta}(u) = \int_\Omega \left[W\left(\frac{x}{\delta}, u\right) + \varepsilon^2 |\nabla u|^2 \right] \mathrm{d}x$$

3 different regimes for Γ-limit:

- $\varepsilon \ll \delta$, Cristoferi, Fonseca, Ganedi (ongoing)
- $\varepsilon \approx \delta$, Cristoferi, Fonseca, Hagerty, Popovicy (2019)
- $\varepsilon \gg \delta$, Cristoferi, Fonseca, Ganedi (2025)

Regime $\varepsilon \gg \delta$: $$F^{(1)}(u) = \sigma^{\mathrm{h}} \operatorname{Per}(\{u = a\}; \Omega) \qquad u \in \mathrm{BV}(\Omega; \{a, b\})$$

$$\sigma^{\mathrm{h}} = \inf \left\{ \int_{-1}^{1} 2\sqrt{W^{\mathrm{h}}(\gamma(t))}\, |\gamma'(t)|\, \mathrm{d}t \right\} \qquad W^{\mathrm{h}}(z) = \int_{Q_1} W(y_1, z)\, \mathrm{d}y_1$$

Two-scale microstructure: $$F_{\varepsilon,\delta,\eta}(u) = \int_\Omega \left[W\left(\frac{x}{\delta}, \frac{x}{\eta}, u\right) + \varepsilon^2 |\nabla u|^2 \right] \mathrm{d}x$$

Separation of scales $\Longrightarrow$ $\eta \ll \delta$ $\Longrightarrow$ 5 different regimes for Γ-limit

$\eta \ll \delta \ll \varepsilon$ $\quad$ $\eta \ll (\delta \approx \varepsilon)$ $\quad$ $\eta \ll \varepsilon \ll \delta$ $\quad$ $(\varepsilon \approx \eta) \ll \delta$ $\quad$ $\varepsilon \ll \eta \ll \delta$

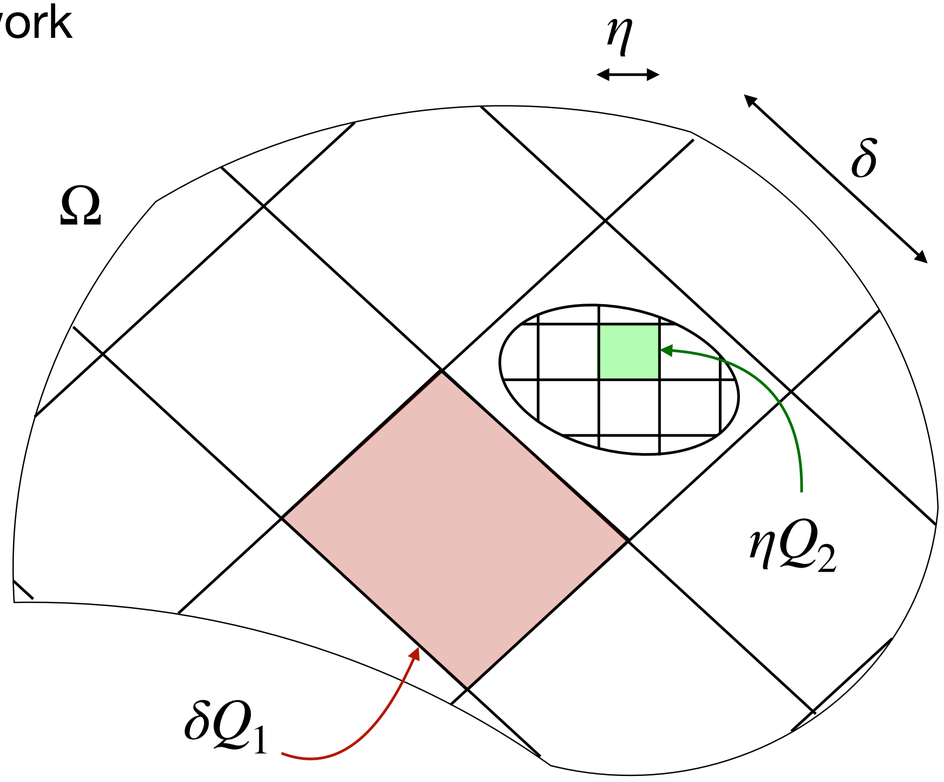
Current work

For simulations, the mesh size has to be at least an order smaller than the smallest physical parameter in the model

$\Longrightarrow$ numerically heavy if parameters tend to 0

$\Longrightarrow$ need for an equivalent functional without parameters

To deal with periodicities, we need **homogenisation**



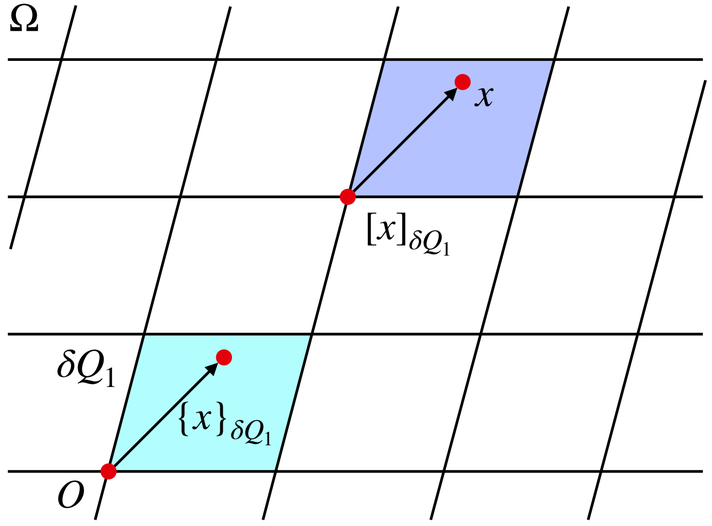
Periodic Unfolding

Cioranescu, Damlamian, Griso (2008, 2022)

Start from identity: $x = \delta \left[\frac{x}{\delta}\right]_{Q_1} + \delta \left\{\frac{x}{\delta}\right\}_{Q_1}$

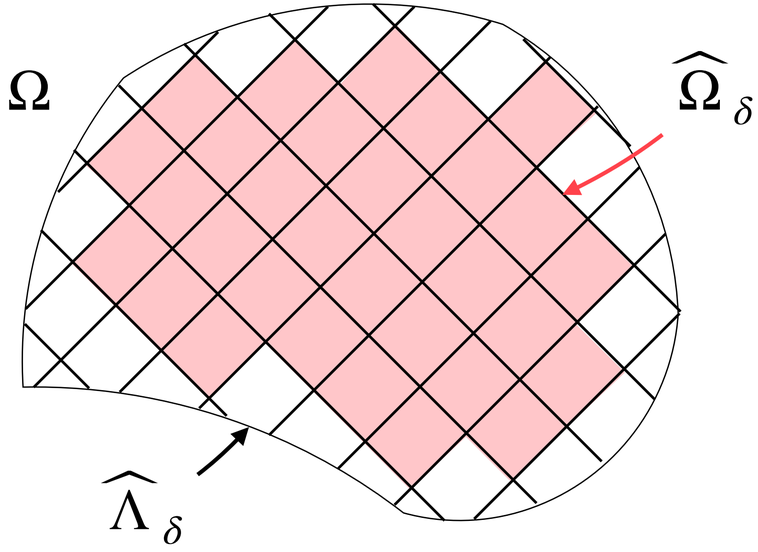
Instead of $\left\{\frac{x}{\delta}\right\}_{Q_1}$ have a variable $y_1 \in Q_1$

$$\Omega \times Q_1 \ni (x, y_1) \longmapsto \delta \left[\frac{x}{\delta}\right]_{Q_1} + \delta y_1 \in \Omega$$



Define **first unfolding operator** $\mathcal{U}_1 \colon L^2(\Omega; \mathbb{R}^M) \to L^2(\Omega; L^2(Q_1; \mathbb{R}^M))$:

$$\mathcal{U}_1 \varphi(x, y_1) = \begin{cases} \varphi\left(\delta \left[\frac{x}{\delta}\right]_{Q_1} + \delta y_1\right) & x \in \widehat{\Omega}_\delta \\ a & x \in \widehat{\Lambda}_\delta \end{cases}$$



Analogously, define **second unfolding operator** $\mathcal{U}_2 \colon L^2(\Omega;\mathbb{R}^M) \to L^2(\Omega; L^2(Q_1; L^2(Q_2;\mathbb{R}^M)))$:

$$\mathcal{U}_2\varphi(x,y_1,y_2) = \begin{cases} \varphi\left(\delta\left[\dfrac{x}{\delta}\right]_{Q_1} + \eta\left[\dfrac{\delta y_1}{\eta}\right]_{Q_2} - \eta\,\iota_{2,\eta} + \eta y_2\right) & (x,y_1) \in \widehat{\Omega}_\delta \times \widehat{Q}_{1,\eta} \\ a & \text{otherwise} \end{cases}$$

where $\iota_{2,\eta} = \left\{\dfrac{\delta}{\eta}\left[\dfrac{x}{\delta}\right]_{Q_1}\right\}_{Q_2} \in Q_2$ accounts for the mismatch between microstructures

Useful properties:

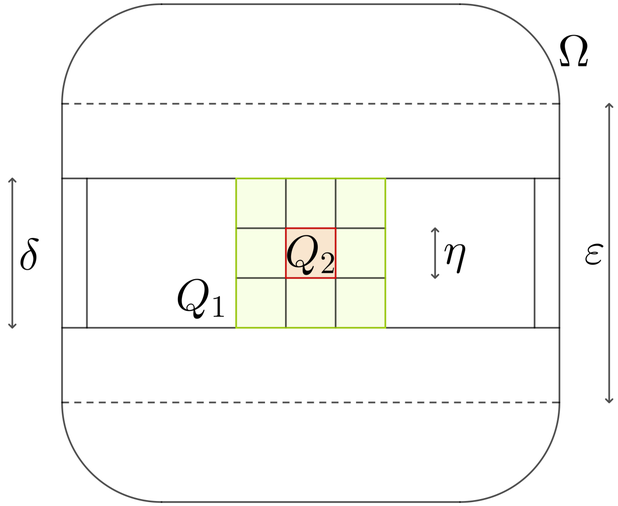
- $\mathcal{U}_1, \mathcal{U}_2$ are linear, bounded and continuous operators
- For $\varphi \in W^{1,2}(\Omega;\mathbb{R}^M)$ the chain rule holds, and moreover

  $\|\nabla_{y_1}\mathcal{U}_1\varphi\|_{L^2} \le \delta\|\nabla\varphi\|_{L^2} \qquad \|\nabla_{y_2}\mathcal{U}_2\varphi\|_{L^2} \le \eta\|\nabla\varphi\|_{L^2}$
- $\displaystyle\int_\Omega W\left(\frac{x}{\delta},\frac{x}{\eta},u\right)\mathrm{d}x \ge \int_\Omega\int_{Q_1}\int_{Q_2} W\left(y_1,y_2,\mathcal{U}_2u\right)\mathrm{d}y_2\mathrm{d}y_1\mathrm{d}x$

Part 3: Main results for regime $\eta \ll \delta \ll \varepsilon$

Assumptions:

- $\Omega \subset \mathbb{R}^N$ bounded open set with Lipschitz boundary
- $W\colon \mathbb{R}^N \times \mathbb{R}^N \times \Omega \to [0, +\infty)$ double well potential with <u>fixed</u> wells in $\{a, b\} \subset \mathbb{R}^M$, Q_1-periodic in y_1, Q_2-periodic in y_2, continuous in z and with super-linear growth at ∞



Main result: Define $F^{(1)}_{\varepsilon_n,\delta_n,\eta_n} = F^{(1)}_n \colon L^1(\Omega;\mathbb{R}^M) \to [0, +\infty]$ as

$$F^{(1)}_n(u) = \begin{cases} \displaystyle\int_\Omega \left[\frac{1}{\varepsilon_n} W\left(\frac{x}{\delta_n}, \frac{x}{\eta_n}, u(x) \right) + \varepsilon_n |\nabla u(x)|^2 \right] \mathrm{d}x & u \in W^{1,2}(\Omega;\mathbb{R}^M) \\ +\infty & u \in L^1(\Omega;\mathbb{R}^M) \setminus W^{1,2}(\Omega;\mathbb{R}^M) \end{cases}$$

When $\eta_n \ll \delta_n \ll \varepsilon_n$ we have

$$\boxed{F^{(1)}_n \xrightarrow{\Gamma(L^1)} F^{(1)}(u) = \begin{cases} \sigma^{\mathrm{h}} \operatorname{Per}(\{u = a\};\Omega) & u \in \mathrm{BV}(\Omega;\{a,b\}) \\ +\infty & \text{otherwise} \end{cases}}$$

where $\sigma^{\mathrm{h}} = \inf \left\{ \displaystyle\int_{-1}^{1} 2\sqrt{W^{\mathrm{h}}(\gamma(t))}\, |\gamma'(t)|\, \mathrm{d}t \right\}$ and $W^{\mathrm{h}}(z) = \displaystyle\int_{Q_1} \int_{Q_2} W(y_1, y_2, z)\, \mathrm{d}y_2 \mathrm{d}y_1$

LIMINF INEQUALITY

$\forall u \in \mathrm{BV}(\Omega; \{a, b\})$ and $\forall \{u_n\}_{n\in\mathbb{N}} \subset W^{1,2}(\Omega; \mathbb{R}^M)$ such that $u_n \longrightarrow u$ strongly in $L^1(\Omega; \mathbb{R}^M)$

it holds that $F^{(1)}(u) \leq \liminf_{n\to\infty} F_n^{(1)}(u_n)$

Strategy:

- Use property of unfolding operator to lower the energy

$$\liminf_{n\to\infty} F_n^{(1)} \geq \liminf_{n\to\infty} \int_\Omega \left[\frac{1}{\varepsilon_n} \int_{Q_1} \int_{Q_2} W\left(y_1, y_2, \mathcal{U}_2 u_n\right) \mathrm{d}y_2 \mathrm{d}y_2 + \varepsilon_n |\nabla u_n|^2 \right] \mathrm{d}x$$

- By writing $\mathcal{U}_2 u_n = u_n + \left[\mathcal{U}_1 u_n - u_n\right] + \left[\mathcal{U}_2 u_n - \mathcal{U}_1 u_n\right]$ we have the idea to define an auxiliary potential $W^\xi \colon \mathbb{R}^M \to [0, +\infty)$ depending on $\xi > 0$ as

$$W^\xi(z) = \inf_{\psi_1 \in \mathcal{A}_1^\xi} \inf_{\psi_2 \in \mathcal{A}_2^\xi} \int_{Q_1} \int_{Q_2} W\left(y_1, y_2, z + \psi_1(y_1) + \psi_2(y_1, y_2)\right) \mathrm{d}y_2 \mathrm{d}y_1$$

such that W^ξ is continuous and $W^\xi \longrightarrow W^{\mathrm{h}}$ uniformly on compact sets as $\xi \to 0$

- Continuity property obtained with

$$\mathscr{A}_1^\xi = \left\{ \psi_1 \in W^{1,2}(Q_1;\mathbb{R}^M) : \|\psi_1\|_{L^2} \le \xi,\ \|\nabla_{y_1}\psi_1\|_{L^2} \le 1 \right\}$$

$$\mathscr{A}_2^\xi = \left\{ \psi_2 \in L^2(Q_1; W^{1,2}(Q_2;\mathbb{R}^M)) : \|\psi_2\|_{L^2} \le \xi,\ \|\nabla_{y_2}\psi_2\|_{L^2} \le 1 \text{ for a.e. } y_1 \in Q_1 \right\}$$

- If we prove that $\left[\mathscr{U}_1 u_n - u_n\right] \in \mathscr{A}_1^\xi \qquad \left[\mathscr{U}_2 u_n - \mathscr{U}_1 u_n\right] \in \mathscr{A}_2^\xi$

  then we have $\displaystyle\int_{Q_1}\int_{Q_2} W\left(y_1, y_2, \mathscr{U}_2 u_n\right) \mathrm{d}y_2 \mathrm{d}y_1 \ge W^\xi(u_n)$

- Young's inequality and a classical result give us

$$\liminf_{n\to\infty} F_n^{(1)}(u_n) \ge \liminf_{n\to\infty} \int_\Omega \sqrt{W^\xi(u_n)}\,|\nabla u_n|\,\mathrm{d}x \ge \sigma^\xi \operatorname{Per}(\{u = a\};\Omega)$$

- Proof that $\sigma^\xi \longrightarrow \sigma^{\mathrm{h}}$ as $\xi \to 0$ based on properties of geodesics

$\blacksquare$

LIMSUP INEQUALITY

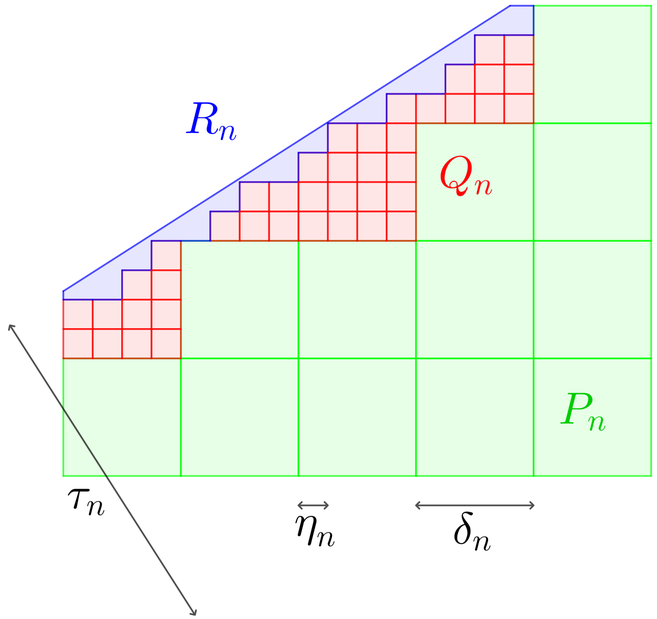
$\forall u \in \mathrm{BV}(\Omega; \{a, b\})$ $\exists \{u_n\}_{n\in\mathbb{N}} \subset W^{1,2}(\Omega; \mathbb{R}^M)$ such that $u_n \longrightarrow u$ strongly in $L^1(\Omega; \mathbb{R}^M)$

and it holds that $F^{(1)}(u) \geq \limsup_{n\to\infty} F_n^{(1)}(u_n)$

Strategy:

- Approximate $A = \{u = a\}$ with C^2 sets, close both in $L^1(\Omega; \mathbb{R}^M)$ and in energy $F^{(1)}(\chi_A)$
- Find optimal curve γ and reparameterisation g_n relative to σ^{h}, and define recovery sequence

$$u_n(x) = \begin{cases} a & \mathrm{dist}(x, \partial A) < -\tau_n \\ \gamma\big(g_n\left(\mathrm{dist}\,(x, \partial A)\right)\big) & -\tau_n \leq \mathrm{dist}(x, \partial A) \leq \tau_n \\ b & \tau_n < \mathrm{dist}(x, \partial A) \end{cases}$$



- Disjointly decompose $A_n = \left\{x \in \Omega : |\,\mathrm{dist}(x, \partial A)\,| \leq \tau_n\right\}$ as in picture and prove that

$$\lim_{n\to\infty} \frac{1}{\varepsilon_n} \int_{A_n} \left| W\left(\frac{x}{\delta_n}, \frac{x}{\eta_n}, u_n\right) - W^{\mathrm{h}}(u_n) \right| \mathrm{d}x = 0$$

- Conclude as in classic Modica-Mortola, with optimal profile properties and coarea formula

$\blacksquare$

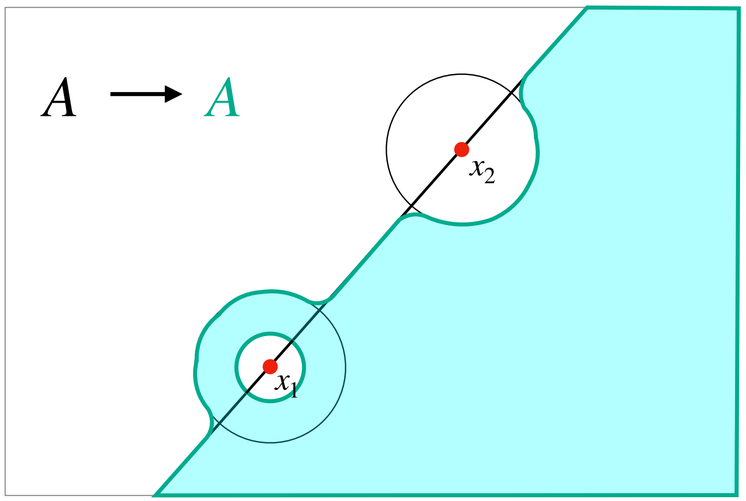
INCORPORATING THE MASS CONSTRAINT

Only need to change the proof of limsup inequality

Step 1: change the approximation of sets such that $|A| = m$

Pick $x_1, x_2 \in \partial A$ and $k \in \mathbb{N}$ big enough and define D as the C^2 approximation of $A \cup B\left(x_1, \frac{1}{k}\right) \backslash B\left(x_2, \frac{1}{k}\right)$

If $|D| > m$ define $A = D \backslash B(x_1, r)$ for an appropriate $r < \frac{1}{k}$

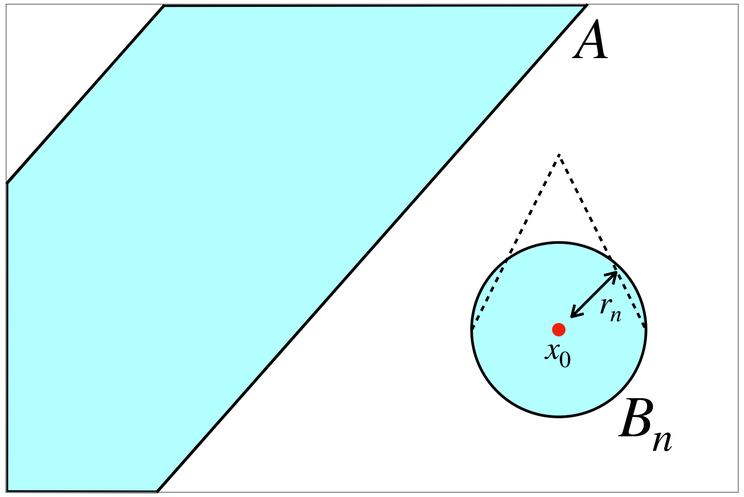


Step 2: change the recovery sequence such that $\int_\Omega u_n \, \mathrm{d}x = m$

Define $m_n = \int_\Omega u_n \, \mathrm{d}x$, pick $r_n = \varepsilon_n^{\frac{2}{1+N}}$, $\quad c_n = -\frac{N+1}{\omega_N} \frac{1}{r_n^N}$

Define $B_n = B(x_0, r_n)$, with $x_0 \in \Omega \backslash A$

$$\tilde{u}_n(x) = \begin{cases} u_n(x) & x \in \Omega \backslash B_n \\ a + c_n(m_n - m)\left(1 - \frac{|x - x_0|}{r_n}\right) & x \in B_n \end{cases}$$



Part 3.5: Main results for regime $\eta \ll \varepsilon \ll \delta$

Assumptions: Identical as first regime, except
$W\colon \mathbb{R}^N \times \mathbb{R}^N \times \Omega \to [0, +\infty)$ locally Lipschitz

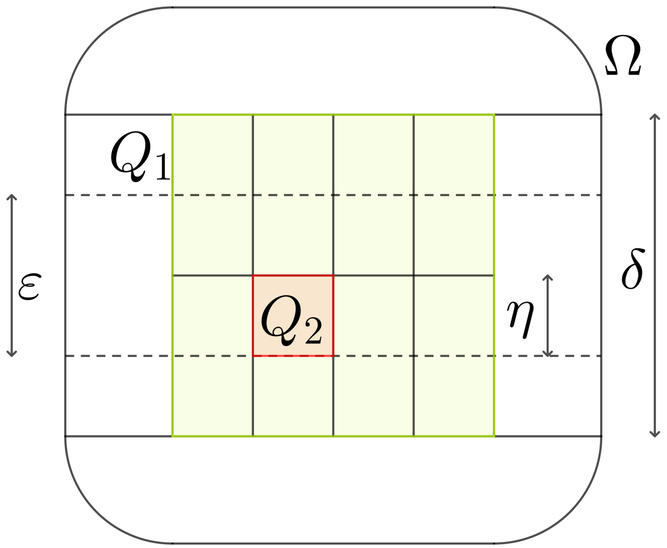
Main result: When $\eta_n \ll \delta_n \ll \varepsilon_n$ we have

$$F_n^{(1)} \xrightarrow{\Gamma(L^1)} F^{(1)}(u) = \begin{cases} \displaystyle\int_{\partial^*\{u=a\}} \sigma\big(\nu(x)\big)\,\mathrm{d}\mathscr{H}^{N-1}(x) & u \in \mathrm{BV}(\Omega;\{a,b\}) \\ +\infty & \text{otherwise} \end{cases}$$

where $\sigma(\nu) = \lim\limits_{T\to+\infty} \dfrac{1}{T^{N-1}} \displaystyle\int_{TQ_\nu \cap \partial E_\nu} \mathrm{d}_\eta^{\mathrm{h}}(y_1)\,\mathrm{d}\mathscr{H}^{N-1}(y_1)$

$$\mathrm{d}_\eta^{\mathrm{h}}(y_1) = \inf\left\{ \int_{-1}^{1} 2\sqrt{W_\eta^{\mathrm{h}}(y_1,\gamma(t))}\,|\gamma'(t)|\,\mathrm{d}t \right\}$$

$$W_\eta^{\mathrm{h}}(y_1,z) = \int_{Q_2} W(y_1,y_2,z)\mathrm{d}y_2$$



LIMINF STRATEGY:

- Define auxiliary potential $W_\eta^\xi(y_1, z)$ such that $W_\eta^\xi \to W_\eta^{\mathrm{h}}$ uniformly as $\xi \to 0$, to get

$$\liminf_{n\to\infty} F_n^{(1)}(u_n) \geq \liminf_{n\to\infty} \int_\Omega 2\sqrt{W_\eta^\xi\left(\frac{x}{\delta_n}, u_n\right)}\,|\nabla u_n|\,\mathrm{d}x$$

- Blow-up and slicing to prove that

$$\lim_{m\to\infty} \int_{Q_\nu} \sqrt{W_\eta^\xi(L_m y_1, v_m(y_1))}\,|\nabla v_m(y_1)|\,\mathrm{d}y_1 \geq \liminf_{n\to\infty} \int_{Q_\nu\cap\partial E_\nu} \mathrm{d}_\eta^\xi(L_m y_1)\,\mathrm{d}\mathscr{H}^{N-1}(y_1)$$

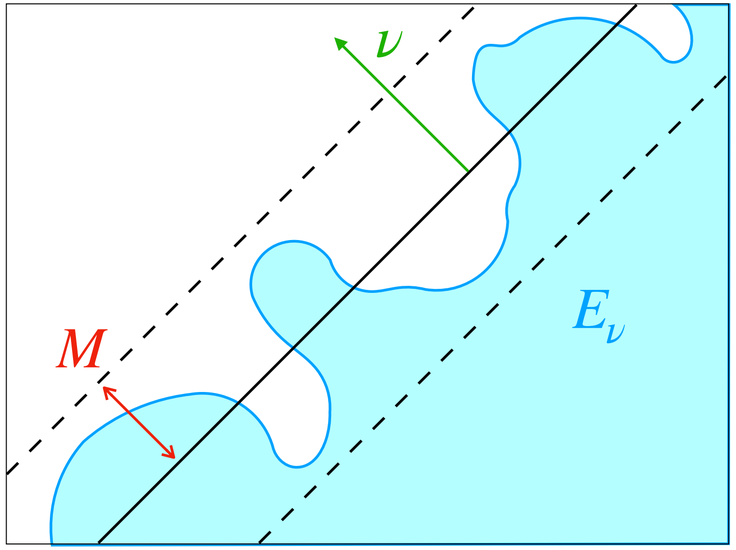
- Prove that $\mathrm{d}_\eta^\xi \longrightarrow \mathrm{d}_\eta^{\mathrm{h}}$ as $\xi \to 0$

LIMSUP STRATEGY:

- Approximate interface and plane-like minimisers with polyhedral sets

- Define recovery sequence with a gluing construction of pointwise optimal profiles

- Prove that contribution of smaller scale is negligible



ONGOING RESEARCH

Other regimes:

$$\varepsilon \ll \eta \ll \delta \qquad (\varepsilon \approx \eta) \ll \delta \qquad \eta \ll (\delta \approx \varepsilon)$$

Sharp interface dynamics:

Study of the convergence of the periodic Allen-Cahn flow

$$\begin{cases} \partial_t u_n = \Delta u_n - \dfrac{1}{\varepsilon_n} \partial_u W\left(\dfrac{x}{\delta_n}, u_n\right) & \text{in } \Omega \times [0,\infty) \\ \nabla u_n \cdot \nu_\Omega = 0 & \text{on } \partial\Omega \times [0,\infty) \end{cases}$$

THANK YOU FOR THE ATTENTION!

FEEL FREE TO ASK ANY QUESTIONS TO ME
(OR TO RICCARDO IN THE AUDIENCE)

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